

Human-language version of the Lean formalisation

O-minimal structures, definable functions, and local properties

1 Ambient ordered set and powers

Definition 1.1 (Dense linear order without endpoints). *A dense linear order without endpoints is a type/set R equipped with a binary relation $<$ such that:*

- (i) $\neg(x < x)$ for every $x \in R$;
- (ii) if $x < y$ and $y < z$, then $x < z$;
- (iii) for all $x, y \in R$, exactly one of $x < y$, $x = y$, or $y < x$ holds;
- (iv) if $x < y$, then there is $z \in R$ with $x < z < y$;
- (v) for every $x \in R$, there are $y, z \in R$ with $y < x < z$.

This is the mathematical content of the Lean structure `DenseLinearOrderNoEndpoints`.

For a natural number n , we write R^n for the set of n -tuples of elements of R . In Lean this is implemented as functions

$$\text{Fin}(n) \rightarrow R,$$

because a tuple of length n is equivalently a function whose domain is the finite set of n indices.

If $x \in R^n$ and $y \in R^m$, their concatenation is written

$$x \frown y \in R^{n+m}.$$

If $z \in R^{n+m}$, then $z_L \in R^n$ and $z_R \in R^m$ denote the left and right coordinate blocks. Thus

$$(x \frown y)_L = x, \quad (x \frown y)_R = y.$$

These are the human versions of the Lean definitions `Power.append`, `Power.left`, and `Power.right`.

For subsets we use the ordinary operations

$$\emptyset, \quad A \cup B, \quad A \cap B, \quad R^n \setminus A.$$

In Lean these are implemented directly as predicates:

$$\emptyset(x) \equiv \perp, \quad (A \cup B)(x) \equiv A(x) \vee B(x), \quad (A \cap B)(x) \equiv A(x) \wedge B(x), \quad A^c(x) \equiv \neg A(x).$$

2 Intervals and one-dimensional definable sets

To state the o-minimality axiom, the Lean file introduces formal endpoints

$$-\infty, \quad a \in R, \quad +\infty.$$

The order on endpoints is the expected one: $-\infty$ is below all finite points, $+\infty$ is above all finite points, and finite endpoints are compared using the given order on R .

For endpoints $a, b \in R \cup \{-\infty, +\infty\}$, the corresponding open interval is

$$(a, b) = \{x \in R : a < x < b\},$$

with the obvious interpretation at infinite endpoints. A point-set is a singleton

$$\{a\} = \{x \in R : x = a\}.$$

Definition 2.1 (Finite unions of points and intervals). *A subset $A \subseteq R$ is a finite union of points and intervals if it belongs to the smallest class containing:*

- (i) *the empty set;*
- (ii) *every singleton $\{a\}$;*
- (iii) *every open interval (a, b) with possibly infinite endpoints;*
- (iv) *the union of any two members of the class.*

This is the inductive predicate `FiniteUnionOfPointsAndIntervals` in the Lean file.

3 O-minimal structures

Definition 3.1 (O-minimal structure). *Let $(R, <)$ be a dense linear order without endpoints. An o-minimal structure on R consists of a family*

$$\mathcal{S}_n \subseteq \mathcal{P}(R^n), \quad n = 0, 1, 2, \dots,$$

satisfying the following closure properties.

- (i) $\emptyset \in \mathcal{S}_n$ for every n .
- (ii) If $A, B \in \mathcal{S}_n$, then $A \cup B \in \mathcal{S}_n$.
- (iii) If $A, B \in \mathcal{S}_n$, then $A \cap B \in \mathcal{S}_n$.
- (iv) If $A \in \mathcal{S}_n$, then $R^n \setminus A \in \mathcal{S}_n$.
- (v) For $i < j < n$, the diagonal

$$\{x \in R^n : x_i = x_j\}$$

belongs to $\mathcal{S}_n$.

- (vi) If $A \in \mathcal{S}_n$ and $B \in \mathcal{S}_m$, then

$$A \times B = \{z \in R^{n+m} : z_L \in A \text{ and } z_R \in B\}$$

belongs to $\mathcal{S}_{n+m}$.

(vii) If $A \in \mathcal{S}_{n+1}$, then each one-coordinate projection of A belongs to $\mathcal{S}_n$.

(viii) Reindexing coordinates preserves definability: if $A \in \mathcal{S}_n$ and $\sigma : \{0, \dots, n-1\} \rightarrow \{0, \dots, m-1\}$, then

$$\{x \in R^m : (x_{\sigma(0)}, \dots, x_{\sigma(n-1)}) \in A\} \in \mathcal{S}_m.$$

(ix) Block existential projection preserves definability: if $A \in \mathcal{S}_{n+m}$, then

$$\{x \in R^n : \exists y \in R^m, x \frown y \in A\} \in \mathcal{S}_n.$$

(x) The order relation

$$\{(x, y) \in R^2 : x < y\}$$

belongs to $\mathcal{S}_2$.

(xi) O-minimality: for every $A \subseteq R$,

$$A \in \mathcal{S}_1 \iff A \text{ is a finite union of points and intervals.}$$

This is the Lean structure `OMinimalStructure`.

Remark 3.2. Some closure properties above, especially reindexing and block existential projection, are often derived from more primitive axioms. They are included explicitly in the Lean file to avoid a large amount of finite-coordinate bookkeeping. All subsequent arguments use only the listed operations.

A set $A \subseteq R^n$ is called *definable* if $A \in \mathcal{S}_n$.

4 Definable functions and composition

Definition 4.1 (Definable function). Let $A \subseteq R^m$ and $B \subseteq R^n$. A function

$$f : A \rightarrow B$$

is definable if $A \in \mathcal{S}_m$, $B \in \mathcal{S}_n$, and its graph

$$\text{Graph}(f) = \{(x, y) \in R^{m+n} : x \in A \text{ and } y = f(x)\}$$

belongs to $\mathcal{S}_{m+n}$.

In the Lean file, the function has type

$$\{x \in R^m : A(x)\} \rightarrow \{y \in R^n : B(y)\},$$

so domain and codomain membership are built into the type of the function. The graph is the predicate

$$z \in \text{Graph}(f) \iff \exists h : A(z_L), z_R = f(z_L, h).$$

Here h is the proof that z_L lies in the domain; it is needed in type theory because the input to f is not merely a point of R^m , but a point together with a proof of membership in A .

Lemma 4.2 (Definability of relational composition). *Let*

$$G \subseteq R^{a+b}, \quad H \subseteq R^{b+c}$$

be definable. Define

$$H \circ G = \{(x, z) \in R^{a+c} : \exists y \in R^b, (x, y) \in G \text{ and } (y, z) \in H\}.$$

Then $H \circ G \in \mathcal{S}_{a+c}$.

Proof. Work in the larger space $R^{(a+c)+b}$ with coordinates written as (x, z, y) . By reindexing coordinates, the set

$$G' = \{(x, z, y) : (x, y) \in G\}$$

is definable. Similarly,

$$H' = \{(x, z, y) : (y, z) \in H\}$$

is definable. Hence $G' \cap H'$ is definable. Project away the last block $y \in R^b$. The resulting subset of R^{a+c} is exactly

$$\{(x, z) : \exists y, (x, y) \in G \text{ and } (y, z) \in H\},$$

which is the relational composition. Therefore it is definable. $\square$

Proposition 4.3 (Composition of definable functions). *Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be definable functions. Then*

$$g \circ f : A \rightarrow C$$

is definable.

Proof. The domain A and codomain C are definable because they are respectively the domain of f and codomain of g . For the graph, observe that

$$(x, z) \in \text{Graph}(g \circ f)$$

if and only if there exists $y \in R^n$ such that

$$(x, y) \in \text{Graph}(f) \quad \text{and} \quad (y, z) \in \text{Graph}(g).$$

Thus

$$\text{Graph}(g \circ f) = \text{Graph}(g) \circ \text{Graph}(f).$$

The right hand side is definable by the previous lemma. Hence $g \circ f$ is definable. $\square$

5 Common definability devices

For one-dimensional variables we write

$$x < y$$

for the order relation on the corresponding coordinates. For n -tuples and coordinate indices $i, j < n$, define:

$$\begin{aligned} \text{dom}_I(i) &= \{z \in R^n : z_i \in I\}, \\ \text{Graph}_G(i, j) &= \{z \in R^n : (z_i, z_j) \in G\}, \\ \text{Lt}(i, j) &= \{z \in R^n : z_i < z_j\}, \\ \text{Eq}(i, j) &= \{z \in R^n : z_i = z_j\}. \end{aligned}$$

If I and G are definable, then all these sets are definable: $\text{dom}_I(i)$ and $\text{Graph}_G(i, j)$ are obtained by reindexing; $\text{Lt}(i, j)$ is obtained by reindexing the definable order relation; and $\text{Eq}(i, j)$ is a diagonal.

We also use the abbreviations

$$\begin{aligned}\text{val}(a, w, b) &= \{z : a < w < b\}, \\ \text{nhd}(c, x, d) &= \{z : c < x < d\}, \\ \text{core}_G(x, a, b, v) &= \{z : (x, v) \in G \text{ and } a < v < b\}.\end{aligned}$$

Each is definable whenever G is definable, since it is built from finite intersections of definable sets.

The set R^n itself is definable, because

$$R^n = R^n \setminus \emptyset.$$

This is the content of the Lean theorem `setUniv_mem`.

6 Continuity points

Let $I \subseteq R$ be definable and let $f : I \rightarrow R$ be definable. Let $G = \text{Graph}(f) \subseteq R^2$.

Definition 6.1 (Continuity at a point, interval form). *For $x \in R$, say that f is continuous at x on I if $x \in I$ and for every interval (a, b) containing $f(x)$ there is an interval (c, d) containing x such that*

$$y \in I \text{ and } c < y < d \implies a < f(y) < b.$$

Equivalently, in graph language:

$$\begin{aligned}x \in I \quad \text{and} \quad \forall a, b, v \quad [G(x, v) \wedge a < v < b \implies \\ \exists c, d \quad (c < x < d \wedge \forall y, w \quad (I(y) \wedge c < y < d \wedge G(y, w) \implies a < w < b))].\end{aligned}$$

This is the predicate `ContinuousAtOnGraph`.

The Lean proof rewrites this universal statement using complements and existential projections. A proposed pair (c, d) fails for fixed (x, a, b, v) precisely when there are y, w such that

$$y \in I, \quad c < y < d, \quad G(y, w), \quad \neg(a < w < b).$$

Define the counterexample set in R^8 , with coordinates

$$(x, a, b, v, c, d, y, w),$$

by

$$C_8 = \{(x, a, b, v, c, d, y, w) : y \in I, c < y < d, G(y, w), \neg(a < w < b)\}.$$

This set is definable by finite intersections and complements. Projecting away (y, w) gives a definable set

$$C_6(x, a, b, v, c, d)$$

expressing that (c, d) is bad. Thus

$$\neg C_6(x, a, b, v, c, d)$$

expresses that (c, d) works.

Now define

$$G_6(x, a, b, v, c, d) \iff G(x, v) \wedge a < v < b \wedge c < x < d \wedge \neg C_6(x, a, b, v, c, d).$$

This is definable. Projecting away (c, d) gives a definable set

$$E_4(x, a, b, v)$$

which says that there exists a working neighbourhood (c, d) for the interval (a, b) around the value v .

A bad continuity witness is a tuple (x, a, b, v) such that

$$G(x, v) \wedge a < v < b \wedge \neg E_4(x, a, b, v).$$

Call this definable set B_4 . Projecting away (a, b, v) gives a definable set $B_1 \subseteq R$ of points where continuity fails. Therefore

$$\text{Cont}(f) = I \cap (R \setminus B_1)$$

is definable.

Theorem 6.2 (Definability of the continuity locus). *Let $f : I \rightarrow R$ be a definable function on a definable set $I \subseteq R$. Then the set of points of I at which f is continuous is definable.*

Proof. The preceding construction writes the discontinuity locus as a finite sequence of operations starting from the definable sets I , G , and $<$: finite intersections, complements, and existential projections. Each operation preserves definability in an o-minimal structure. Hence the bad set B_1 is definable. Its complement is definable, and intersecting with I is definable. Therefore the continuity locus is definable. $\square$

7 Locally constant points

Definition 7.1 (Local constancy at a point). *The function f is locally constant at x on I if $x \in I$ and there exists an interval (c, d) containing x such that for every $y \in I$,*

$$c < y < d \implies f(y) = f(x).$$

In graph language this is:

$$x \in I \quad \text{and} \quad \exists c, d, v [G(x, v) \wedge c < x < d \wedge \forall y, w (I(y) \wedge c < y < d \wedge G(y, w) \Rightarrow w = v)].$$

This is the predicate `LocallyConstantAtOnGraph`.

For coordinates

$$(x, c, d, v, y, w) \in R^6,$$

define the bad set

$$C_6^{\text{lc}} = \{(x, c, d, v, y, w) : y \in I, c < y < d, G(y, w), w \neq v\}.$$

This is definable: the inequality $w \neq v$ is the complement of the diagonal $w = v$. Projecting away (y, w) gives a definable set

$$C_4^{\text{lc}}(x, c, d, v)$$

which says that a counterexample to constancy exists in the neighbourhood. Hence

$$\neg C_4^{\text{lc}}(x, c, d, v)$$

says that the neighbourhood works.

Define the good set in R^4 by

$$G_4^{\text{lc}}(x, c, d, v) \iff G(x, v) \wedge c < x < d \wedge \neg C_4^{\text{lc}}(x, c, d, v).$$

This is definable. Projecting away (c, d, v) gives a definable subset

$$G_1^{\text{lc}} \subseteq R$$

of points admitting such a neighbourhood and value. The locally constant locus is

$$\text{LC}(f) = I \cap G_1^{\text{lc}}.$$

Theorem 7.2 (Definability of the locally constant locus). *Let $f : I \rightarrow R$ be definable. Then the set of points of I at which f is locally constant is definable.*

Proof. The set C_6^{lc} is built from I , G , the order relation, and the complement of a diagonal, using finite intersections. It is therefore definable. Projection gives C_4^{lc} , complement gives the condition that no counterexample exists, another finite intersection gives G_4^{lc} , and projection gives G_1^{lc} . Finally $I \cap G_1^{\text{lc}}$ is definable. $\square$

8 Locally monotone points

We use the non-strict convention:

locally increasing = locally nondecreasing, locally decreasing = locally nonincreasing.

Definition 8.1 (Local monotonicity at a point). *The function f is locally increasing at x if $x \in I$ and there exists an interval (c, d) containing x such that for all $y_0, y_1 \in I$,*

$$c < y_0 < d, \quad c < y_1 < d, \quad y_0 < y_1 \implies f(y_0) \leq f(y_1).$$

Since we only have the strict order symbol, the conclusion is written as

$$\neg(f(y_1) < f(y_0)).$$

In graph language, for values $w_0 = f(y_0)$ and $w_1 = f(y_1)$, the forbidden inequality is $w_1 < w_0$.

*Similarly, f is locally decreasing at x if the forbidden inequality is $w_0 < w_1$. These are the predicates **LocallyIncreasingAtOnGraph** and **LocallyDecreasingAtOnGraph**.*

Use coordinates

$$(x, c, d, y_0, y_1, w_0, w_1) \in R^7.$$

First define the common base counterexample set

$$C_7^{\text{base}} = \{(x, c, d, y_0, y_1, w_0, w_1) : y_0 \in I, y_1 \in I, c < y_0 < d, c < y_1 < d, \\ G(y_0, w_0), G(y_1, w_1), y_0 < y_1\}.$$

This is definable by the closure properties.

The increasing-bad set is

$$C_7^{\text{inc}} = C_7^{\text{base}} \cap \{w_1 < w_0\}.$$

The decreasing-bad set is

$$C_7^{\text{dec}} = C_7^{\text{base}} \cap \{w_0 < w_1\}.$$

Both are definable.

Projecting away (y_0, y_1, w_0, w_1) gives definable sets

$$C_3^{\text{inc}}(x, c, d), \quad C_3^{\text{dec}}(x, c, d),$$

which say that a bad pair exists in the neighbourhood (c, d) . Thus

$$\neg C_3^{\text{inc}}(x, c, d)$$

says that (c, d) witnesses local increasingness, and

$$\neg C_3^{\text{dec}}(x, c, d)$$

says that (c, d) witnesses local decreasingness.

Define

$$\begin{aligned} G_3^{\text{inc}}(x, c, d) &\iff c < x < d \wedge \neg C_3^{\text{inc}}(x, c, d), \\ G_3^{\text{dec}}(x, c, d) &\iff c < x < d \wedge \neg C_3^{\text{dec}}(x, c, d). \end{aligned}$$

These are definable. Projecting away (c, d) gives definable subsets

$$G_1^{\text{inc}}, \quad G_1^{\text{dec}}$$

of R . The final loci are

$$\text{LInc}(f) = I \cap G_1^{\text{inc}}, \quad \text{LDec}(f) = I \cap G_1^{\text{dec}}.$$

Theorem 8.2 (Definability of local monotonicity loci). *Let $f : I \rightarrow R$ be definable. Then both sets*

$$\{x \in I : f \text{ is locally increasing at } x\}$$

and

$$\{x \in I : f \text{ is locally decreasing at } x\}$$

are definable.

Proof. The common base counterexample set is obtained from I , G , and $<$ using only reindexing and finite intersections. Intersecting it with either $w_1 < w_0$ or $w_0 < w_1$ gives the increasing-bad and decreasing-bad sets. Each is definable. Projection gives the sets of triples (x, c, d) for which a bad pair exists. Taking complements expresses that no bad pair exists. Intersecting with $c < x < d$ and then projecting away (c, d) gives the points admitting a good neighbourhood. Finally intersecting with I gives the desired loci. Each operation preserves definability, so both loci are definable. $\square$

9 Small sanity checks

The Lean file ends with two simple definability checks:

- (i) the empty set is definable in every arity, by the axiom $\emptyset \in \mathcal{S}_n$;
- (ii) the order relation $\{(x, y) : x < y\}$ is definable, by the distinguished axiom `lt.mem`.

10 Dictionary between the Lean file and this document

Lean name	Mathematical object
<code>Power R n</code>	R^n
<code>Power.append</code>	tuple concatenation $x \frown y$
<code>Power.left</code> , <code>Power.right</code>	left and right coordinate blocks
<code>setInter</code> , <code>setUnion</code> , <code>setCompl</code>	$\cap$, $\cup$, complement
<code>DenseLinearOrderNoEndpoints</code>	dense linear order without endpoints
<code>OMinimalStructure</code>	family $(\mathcal{S}_n)_n$ with closure axioms
<code>FunctionGraph</code>	graph of a definable function
<code>relation_comp_mem</code>	relational composition preserves definability
<code>DefinableFunction.comp</code>	composition of definable functions
<code>ContinuousPoints</code>	continuity locus
<code>LocallyConstantPoints</code>	local constancy locus
<code>LocallyIncreasingPoints</code>	locally nondecreasing locus
<code>LocallyDecreasingPoints</code>	locally nonincreasing locus